



mongoDB® Atlas

[MongoDB Atlas](https://cloud.mongodb.com)

Introducción

MongoDB Atlas es la plataforma de base de datos como servicio (DBaaS) basada en la nube de MongoDB Inc.. Permite desplegar, gestionar y escalar bases de datos MongoDB en infraestructuras de nube pública como Amazon Web Services, Google Cloud Platform y Microsoft Azure. Está diseñada para simplificar las operaciones, la seguridad y la escalabilidad de las aplicaciones de datos modernos.

Introducción

Datos clave

- **Lanzamiento:** 2016
- **Fundador:** MongoDB Inc.
- **Tipo:** Servicio DBaaS gestionado en la nube
- **Compatibilidad:** AWS, Azure y Google Cloud
- **Modelo de precios:** Pago por uso con planes gratuitos y escalables

Introducción

Funcionalidades principales

MongoDB Atlas automatiza tareas críticas como aprovisionamiento, actualización, copias de seguridad y monitorización de bases de datos. Integra herramientas para análisis en tiempo real, seguridad avanzada (cifrado, control de acceso granular) y alta disponibilidad mediante replicación automática y distribución geográfica de datos.

Introducción

Seguridad y cumplimiento

El servicio ofrece cifrado de datos en reposo y en tránsito, autenticación multifactor y cumplimiento con normativas como GDPR, SOC 2 y HIPAA. Los usuarios pueden definir políticas de acceso basadas en roles y configurar redes privadas virtuales para aislar entornos.

Introducción

Escalabilidad y rendimiento

Atlas permite escalar vertical y horizontalmente mediante particionado (“sharding”) y ajuste automático de recursos. Su arquitectura distribuida soporta cargas de trabajo globales, con réplicas regionales para reducir la latencia y aumentar la resiliencia.

Introducción

Integraciones y ecosistema

- El servicio se integra con el ecosistema de MongoDB Compass, MongoDB Charts y herramientas de terceros para DevOps, análisis y visualización. Además, ofrece APIs y SDKs para múltiples lenguajes, facilitando la conexión de aplicaciones web y móviles con datos alojados en la nube.
- MongoDB Atlas se ha consolidado como una solución clave para empresas que buscan una base de datos NoSQL gestionada y segura en entornos multinube.

Document databases

Stored in documents

Document databases use JSON documents

Data stored in key-value pairs - flexible!

Because of the flexibility of document databases, we can store related data in the same document.

```
{  
  "key": "value"  
}
```

```
{  
  "_id": 1,  
  "name": "Jesse",  
  "phone": [  
    "333-3333", "444-4444"  
  ]  
}
```

Relational

database

table

row

column

MongoDB

database

collection

document

field

Select

```
SELECT * FROM people
```

```
SELECT * FROM people  
WHERE age=25
```

```
SELECT * FROM people  
WHERE age>25
```

Find

```
db.people.find({})
```

```
db.people.find({ age: 25 })
```

```
db.people.find({  
  age: { $gt: 25 }  
})
```

SELECT ... FROM

→ find()

WHERE

→ filtro {campo: condición}

ORDER BY

→ .sort()

LIMIT

→ .limit()

GROUP BY

→ aggregate() con \$group (+ \$match, \$sort, etc.)

Filtros (WHERE)

Operadores típicos:

- \$gt, \$gte, \$lt, \$lte (comparaciones)
- \$in (pertenencia)
- \$regex (patrones)
- \$and, \$or (lógicos)



ORGANIZATION
CSIC ▼



PROJECT
Project 0 ▼



Deploy your cluster

Use a template below or set up advanced configuration options. You can also edit these configuration options once the cluster is created.

☐ M10

\$0.09/hour

Dedicated cluster for development environments and low-traffic applications.

STORAGE

10 GB

RAM

2 GB

vCPU

2 vCPUs

☐ Flex

From \$0.011/hour

Up to \$30/month

For development and testing, with on-demand burst capacity for unpredictable traffic.

STORAGE

5 GB

RAM

Shared

vCPU

Shared

☒ Free

For learning and exploring MongoDB in a cloud environment.

STORAGE

512 MB

RAM

Shared

vCPU

Shared



Privacidad - Términos

I'll do this later

Go to Advanced Configuration

Create Deployment



Client drivers

 C

 C++

 C#

 Go

 Java

 Node.js

 Perl

 PHP

 Python

 Ruby

 Scala

 Swift

 Rust

 R

Project Overview



DATABASE



Clusters

Search & Vector Search

Data Explorer

Backup

STREAMING DATA



Stream Processing

Triggers

SERVICES



AI Module

Clusters

Find a database deployment...

Cluster0

Connect

View Monitoring

Browse Collections

...

- Edit Configuration
- Command Line Tool
- Load Sample Database
- Terminate

Visualize Your Data

Build dashboards and charts, and embed them in your apps with MongoDB Charts.

Explore Charts

Dismiss

R 0

W

Last 15 minutes

100.0%

Load sample data

Select sample datasets

sample_airbnb ✕

sample_restaurants ✕



Cancel

Load sample data

Project Overview

DATABASE

Clusters

Search & Vector Search

Data Explorer

Backup

STREAMING DATA

Stream Processing

Triggers

SERVICES

AI Models

Migration

Data Federation

Visualization

SECURITY

Clusters

 Edit Config ▾

● Cluster0

Connect

View Monitoring

Browse Collections

...

Visualize Your Data

Build dashboards and charts, and embed them in your apps with MongoDB Charts.

[Explore Charts](#)[Dismiss](#)

● R 0

● W

Last 7 minutes

100.0/s



Connections 2.0

Last 7 minutes

2.0



● In 324.17 KB/s

● Out 1.95 KB/s

Last 7 minutes

324.17 KB/s



Data In

115.7

KB/s

512.001

VERSION

REGION

TYPE

BACKUPS

LINKED APP SERVICES

ATLAS SQL

ATLAS SEARCH

8.0.19

AWS / Paris (eu-west-3)

Replica Set - 3 nodes

Inactive

None Linked

[Connect](#)[1 search index](#)[+ Add Tag](#)

Connect to Cluster0



You need to secure your MongoDB Atlas cluster before you can use it. Set which users and IP addresses can access your cluster now. [Read more](#)

1. Add a connection IP address

✓ Your current IP address (79.117.229.93) has been added to enable local connectivity. Only an IP address you add to your Access List will be able to connect to your project's clusters. Add more later in [Network Access](#).

2. Create a database user

This first user will have [atlasAdmin](#) permissions for this project.

i You'll need your database user's credentials in the next step. Copy the database user password.

Username

nebrija_profe

Password

pass_nebrijd

HIDE

Copy

Create Database User

Close

Choose a connection method

Connect to Cluster0



Connect to your application



Drivers

Access your Atlas data using MongoDB's native drivers (e.g. Node.js, Go, etc.)



Access your data through tools



Compass

Explore, modify, and visualize your data with MongoDB's GUI



Shell

Quickly add & update data using MongoDB's Javascript command-line interface



MongoDB for VS Code

Work with your data in MongoDB directly from your VS Code environment



Atlas SQL

Easily connect SQL tools to Atlas for data analysis and visualization



Go Back

Close

Connect to Cluster0



Connecting with MongoDB Driver

1. Select your driver and version

We recommend installing and using the latest driver version.

Driver	Version
Python	3.12 or later

2. Install your driver

Run the following on the command line

Note: Use appropriate Python 3 executable

```
python -m pip install "pymongo[srv]>=3.12"
```

[View MongoDB Python Driver installation instructions.](#)

3. Add your connection string into your application code

Use this connection string in your application

☐ View full code sample

```
mongodb+srv://nebrija_profe:<db_password>@cluster0.hbmynut.mongodb.net/?  
appName=Cluster0
```

Replace `<db_password>` with the password for the `nebrija_profe` database user. Ensure any option params are [URL encoded](#).

RESOURCES

[Get started with the Python Driver](#)

[Python Starter Sample App](#)

[Access your Database Users](#)

[Troubleshoot Connections](#)

Go Back

Done

Project Overview

DATABASE

Clusters

Search & Vector Search

Data Explorer

Backup

STREAMING DATA

Stream Processing

Triggers

SERVICES

AI Models

Migration

Data Federation

Visualization

SECURITY

Security Quickstart

Project Identity & Access

Database & Network Access

Project 0 Overview

Clusters

[Create cluster](#)

Cluster0



Sample data and search index loaded successfully. Get started with a sample query in [Data Explorer](#) or by connecting with MongoDB Shell.

[Connect](#)[Edit configuration](#)

Data Size: 220.55 MB

[Browse collections](#)[View monitoring](#)[+ Add Tag](#)

Application Development

[Get connection string](#)

Python



Connect your application to your database deployments with the official MongoDB libraries.

[Start with a Python Sample App](#)

Toolbar

[Resources](#)[Tips \(1\)](#)

Performance (1)



Cost (0)



Resilience (0)



IDENTITY & ACCESS

All Projects

Users

Applications

Teams

Federation

AI MODELS

Model API Keys

Usage

Rate Limits

BILLING

Overview

Invoices

Cost Explorer

CONFIGURATIONS

Resource Policies

Activity Feed

Organization Settings

ORGANIZATION

CSIC

?

All Projects

New Project

Find a project...

Project Name	Clusters	Tags	Users	Teams	Alerts	Actions
Project 0	1 Cluster	+ Add Tags	2 Users	0 Teams	0 Alerts	

System Status: All Good

Last Login: 79.117.229.93

movies



Cluster0 > sample_mflix > movies

View monitoring

Visualize Your Data

Documents 21K

Aggregations

Schema

Indexes 2

Validation



Your pipeline is currently empty.

Edit



Explain

Run

Options

Collation

movies.find_one()

Max Time MS

60000

ALL RESULTS

Showing 1 – 20 count results



```
_id: ObjectId('573a1390f29313caabcd42e8')
plot: "A group of bandits stage a brazen train hold-up, only to find a determ..."
  genres: Array (2)
  runtime: 11
  cast: Array (4)
  poster: "https://m.media-amazon.com/images/M/MV5BMTU3NjE5NzYtYTUyNS00MDVmLWIwYj..."
  title: "The Great Train Robbery"
  fullplot: "Among the earliest existing films in American cinema, notable as the "
```


Export Pipeline To Language

My Pipeline

Shell

```
[
  {
    $group: {
      _id: "$bedrooms",
      count: {
        $sum: 1
      }
    }
  },
]
```

☐ Include Import Statements

☐ Include Driver Syntax

Exported Pipeline

Python

```
[
  {
    '$group': {
      '_id': '$bedrooms',
      'count': {
        '$sum': 1
      }
    }
  }, {
```

Close